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Performing data interactions in a virtual environment

Abstract

A system includes a memory and a processor coupled to the memory. The processor receives a first user credential associated with a first user and authorizes a first avatar of the first user to enter a virtual environment. The processor receives a first request from the first user to receive virtual data objects for at least one target entity and, in response, generates a virtual sub-environment within the virtual environment. The processor authorizes a second user to access the virtual sub-environment and receives a set of conditions from the second user relating to how a first set of virtual data objects is to be distributed to the at least one target entity. The processor receives and transfers the first set of virtual data objects to the at least one target entity based on the set of conditions.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to network communication, and more specifically to performing data interactions in a virtual environment.

BACKGROUND

(2) In a network environment, user devices are in data communication with other user devices that may be distributed anywhere in the world. These network environments allow data and information to be shared among these devices. Some of the technical challenges that occur when data is exchanged between devices are controlling data leakage, unauthorized access to data, and preventing malicious activities. Data storing user devices, such as computers, laptops, augmented reality devices, virtual reality devices, and smartphones, are vulnerable to attacks. This vulnerability poses several network security challenges. Existing systems are typically unable to detect a malicious attack until after the attack has occurred. For example, a bad actor may pretend to be another user in a virtual environment which then allows the bad actor to gain access to other users' information.

SUMMARY

(3) The system and methods implemented by the system as disclosed in the present disclosure provide technical solutions to the technical problems discussed above by allowing a user to securely access a virtual environment and perform secure data interactions in the virtual environment. The disclosed system and methods provide several practical applications and technical advantages.

(4) For example, the disclosed system and methods provide the practical application of improving interoperability of real-world systems and virtual world systems (e.g., metaverse systems) so that information may be seamlessly shared between these systems to implement data security, authorization of data interactions, access to virtual sub-environments and other data interactions performed in real-world and virtual environments. For example, user information collected from the user and/or assigned to the user in a real-world environment may be used in a virtual environment (e.g., metaverse environment) to authenticate the user before allowing the user to access the virtual environment and perform any kind of action or interaction within the virtual environment. Additionally or alternatively, as described in embodiments of the present disclosure, user information collected from the user and/or assigned to the user in the real-world environment may be used in the virtual environment (e.g., metaverse environment) to provide access to the user to a virtual sub-environment within the virtual environment. This process provides improved information security because it authenticates that an avatar is associated with the user and not an unauthorized party and that the user is authorized to access the virtual environment and the virtual sub-environment. By improving interoperability of real-world systems and virtual-world systems, the disclosed system and methods generally improve computer networking technology.

(5) The disclosed system and methods provide the additional practical application of recording data interactions performed in the virtual environment and/or the real-world environment in a blockchain. Blockchain supports immutability which means that data transactions once recorded in the blockchain cannot be erased, changed, or replaced. The immutability of the blockchain significantly reduces the chances of data tampering within the network and makes the system significantly secure as compared to traditional client-server architecture. The blockchain further helps ensure traceability and accountability in the event of an unauthorized and malicious activity.

(6) Thus, the disclosed system and methods improve data security in the virtual environment. By improving data security in virtual environment, the disclosed system and methods generally improve technology related to performing secure data interactions in a virtual environment.

(7) The disclosed system and methods provide the additional practical application of saving memory resources. The seamless data flow between the real-world systems and virtual-world systems as a result of interoperability of these systems allows each system to store less data by

avoiding the need to store the same data (e.g., authentication data, login credentials, etc.) in both systems, as data stored in one system can be accessed, or otherwise leveraged, by the other system. This saves memory resources by avoiding duplication of data. The saving of memory resources may leave more system memory for storing other data used in other technical operations. This provides the additional technical advantage of improving processing efficiency of computing systems that manage the real-world and virtual word environments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.
- (2) FIG. 1 is a schematic diagram of a system, in accordance with certain aspects of the present disclosure;
- (3) FIG. 2 is a block diagram of an embodiment of the first user device used by the system of FIG. 1; and
- (4) FIG. 3 illustrates a flowchart of an example method for performing data interactions in a virtual environment, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(5) Example System

(6) FIG. 1 is a schematic diagram of a system **100**, in accordance with certain aspects of the present disclosure. System **100** may include a first user device **104**, a second user device **106**, real-world server **130**, virtual-world server **150** and a blockchain network **190**, each connected to a network **180**. A first user **110** is associated with the first user device **104** and a second user **112** is associated with the second user device **106**. The system **100** may be communicatively coupled to the communication network **180** and may be operable to transmit data between each one of the first user device **104**, second user device **106**, real-world server **130**, virtual-world server **150**, and blockchain network **190** through the communication network **180**.

(7) In general, the system **100** may improve interoperability of real-world systems and virtual world systems (e.g., metaverse systems) so that information may be seamlessly shared between these systems to implement data security, authorization of data interactions, access to virtual sub-environments and other data interactions performed in real-world and virtual environments. For example, user information collected from the user and/or assigned to the user in a real-world environment may be used in a virtual environment **102** (e.g., metaverse environment) to authenticate the first user **110** before allowing the first user **110** to access the virtual environment **102** and perform any kind of action or interaction within the virtual environment **102**. Additionally or alternatively, as described in embodiments of the present disclosure, user information collected from the first user **110** and/or assigned to the first user **110** in the real-world environment may be used in the virtual environment **102** (e.g., metaverse environment) to provide access to the first user **110** to a virtual sub-environment **120** within the virtual environment **102**. This process provides improved information security because it authenticates that a first avatar **114** is associated with the first user **110**, not an unauthorized party, and that the first user **110** is authorized to access the virtual environment **102** and the virtual sub-environment **120**. Additionally or alternatively, as described in embodiments of the present disclosure, data interactions are performed in the real-world environment by the real-world server **130** based on data interactions performed in the virtual environment by the virtual-world server **150**. Further, information relating to data interactions performed in the real-world environment by the real-world server **130** is communicated to the virtual-world server **150** for presenting to the first user **110** in the virtual environment **102** or virtual

sub-environment **120** and for recording in a blockchain network **190**.

(8) It may be noted that the terms “real-world” and “real-world environment” in this disclosure refer to any non-virtual environment where users (e.g., users **110** and **112**) can physically interact with real persons and objects. A real-world data interaction may refer to any data interaction performed outside the virtual environment **102** (e.g., a metaverse environment). Further, it may be noted that while certain embodiments of the present disclosure may be described in the context of a metaverse environment which is an example of a virtual environment **102**, the methods discussed in this disclosure apply to any other virtual environment **102**. The terms “virtual environment” and “metaverse environment” are used interchangeably throughout this disclosure. Furthermore, it may be noted that while certain embodiments of this disclosure describe one or more operations in relation to the first user **110**, these embodiments apply to any other user (e.g., second user **112**) connected to network **180**.

(9) The first user **110** may access the virtual environment **102** (e.g., a metaverse environment) through the first user device **104**. The first user device **104** is configured to display a two-dimensional (2D) or three-dimensional (3D) representation of the virtual environment **102** to the first user **110**. Examples of a virtual environment **102** may include, but are not limited to, a graphical or virtual representation of a metaverse, a map, a building interior, a landscape, a fictional location, an alternate reality, or any other suitable type of location or environment. The virtual environment **102** may be configured to use realistic or non-realistic physics for the motion of objects within the virtual environment **102**. For example, some virtual environments **102** may be configured to use gravity whereas other virtual environments **102** may not be configured to use gravity. Within the virtual environment **102**, each user may be associated with an avatar (such as the first avatar **114** for the first user **110**). An avatar is a graphical representation of a user at a virtual location within the virtual environment **102**. In embodiments, the virtual location of the avatar may be correlated to the physical location of a user in the real-world environment. Examples of an avatar may include, but are not limited to, a person, an animal, or an object. In some embodiments, the features and characteristics of the avatar may be customizable and user-defined. For example, the size, shape, color, attire, accessories, or any other suitable type of appearance features may be specified by a user. By using an avatar, a user is able to move within the virtual environment **102** to interact with one or more avatars and objects within the virtual environment **102** while independently remaining at a physical location in the real-world environment or being in transit in the real-world environment.

(10) While engaging in the virtual environment **102** via the first avatar **114**, the first user **110** may interact with a plurality of other users, objects and/or entities (e.g., virtual sub-environment **120**) through a respective avatar. For example, the second user **112** may attempt to engage in an interaction session with the first avatar **114** through a second avatar **116** associated with the second user **112**. In another example, the first avatar **114** of the first user **110** may access a virtual sub-environment **120** within the virtual environment **102** and perform virtual data interactions within the virtual sub-environment **120**. In the real-world environment, the second user **112** may be physically located at a distance away from the first user **110**. The second user **112** may access the virtual environment **102** through the second user device **106** to control the second avatar **116** and attempt to engage in an interaction session with the first user **110** through the first avatar **114**.

(11) Before the interaction between the first avatar **114** and the second avatar **116** occurs or the first avatar **114** can access the virtual sub-environment **120**, the virtual-world server **150** may authenticate that the first avatar **114** is associated with the first user **110** and not an unauthorized third-party. For example, the first user **110** may be required to sign into a secure portal that provides access to a data file (e.g., real-world data file **134** and/or virtual data file **160**) associated with the first user **110**. As shown in FIG. 1, the real-world data file **134** of the first user **110** is stored and managed by the real-world server **130** and the virtual data file **160** of the first user **110** is stored and managed by the virtual-world server **150**. In one or more embodiments, the virtual-

world server **150** may employ single sign-on (SSO), multifactor authentication, or any other suitable authentication scheme in order to allow the first user **110** access to the virtual data file **160** and/or the real-world data file **134**. The virtual data file **160** and the real-world data file **134** may include virtual data objects **162** and real-world data objects **136** respectively owned by the first user **110**. The real-world server **130** and the virtual-world server **150** may store other information related to the first user **110** including, but not limited to, user profile information, account information (e.g., including identity and other details relating to the respective data files **134** and **160**), avatar information, digital assets (e.g., respective real-world data objects **136** and virtual data objects **162**) information, or any other suitable type of information that is associated with a user within the virtual environment **102** and/or the real-world environment.

(12) As shown in FIG. 1, virtual-world server **150** comprises a processor **170**, a memory **174**, and a network interface **172**. The processor **170** comprises one or more processors operably coupled to the memory **174**. The processor **170** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g. a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **170** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **170** is communicatively coupled to and in signal communication with the memory **174**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **170** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **170** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(13) The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions (e.g., virtual-world server instructions **176**) to implement the virtual-world server **150**. In this way, processor **170** may be a special-purpose computer designed to implement the functions disclosed herein. In one or more embodiments, the virtual-world server **150** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The virtual-world server **150** is configured to operate as described with reference to FIGS. 1 and 3. For example, the processor **170** may be configured to perform at least a portion of the method **300** as described in FIG. 3.

(14) The memory **174** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **174** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(15) The memory **174** is operable to store first user credential **152**, second user credential **154**, virtual security tokens **156**, conditions **158**, virtual data file **160**, virtual data objects **162**, and the virtual-world server instructions **176**. The virtual-world server instructions **176** may include any suitable set of instructions, logic, rules, or code operable to execute the virtual-world server **150**.

(16) The network interface **172** is configured to enable wired and/or wireless communications. The network interface **172** is configured to communicate data between the virtual-world server **150** and other devices, systems, or domains (e.g. user devices **104** and **106**, real-world server **130** and the blockchain network **190**). For example, the network interface **172** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **170** is configured to send and receive data using the network interface **172**. The network interface **172** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(17) The real-world server **130** comprises a processor **140**, a memory **144**, and a network interface **142**. The processor **140** comprises one or more processors operably coupled to the memory **144**. The processor **140** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g. a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **140** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **140** is communicatively coupled to and in signal communication with the memory **144**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **140** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **140** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(18) The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions (e.g., real-world server instructions **146**) to implement the real-world server **130**. In this way, processor **140** may be a special-purpose computer designed to implement the functions disclosed herein. In one or more embodiments, the real-world server **130** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The real-world server **130** is configured to operate as described with reference to FIGS. **1** and **3**. For example, the processor **140** may be configured to perform at least a portion of the method **300** as described in FIG. **3**.

(19) The memory **144** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **144** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(20) The memory **144** is operable to store information relating to user data **132**, real-world data file **134**, real-world data objects **136**, first user credential **152**, second user credential **154** and the real-world server instructions **146**. The real-world server instructions **146** may include any suitable set of instructions, logic, rules, or code operable to execute the real-world server **130**.

(21) The network interface **142** is configured to enable wired and/or wireless communications. The network interface **142** is configured to communicate data between the real-world server **130** and other devices, systems, or domains (e.g. user devices **104** and **106**, virtual-world server **150** and blockchain network **190**). For example, the network interface **142** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **140** is configured to send and receive data using the network interface **142**. The network interface **142** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(22) Each of the real-world server **130** and the virtual-world server **150** is generally a suitable server (e.g., including a physical server and/or virtual server) operable to store data in a memory (e.g., respective memories **144** and **174**) and/or provide access to application(s) or other services. One or both of the real-world server **130** and the virtual-world server **150** may be a backend server associated with a particular entity (e.g., organization) that facilitates conducting interactions between entities and one or more users. In other embodiments, one or both of the real-world server **130** and the virtual-world server **150** may be organized in a distributed manner, or by leveraging cloud computing technologies. Real-world server **130** may store information which is primarily used to support data interactions performed in the real-world environment. Virtual-world server **150** may store information which is primarily used to support data interactions performed in the

virtual environment **102** (e.g., a metaverse environment). It may be noted that the operations performed by the real-world server **130** and the virtual-world server **150** described in embodiments of the present disclosure may be implemented by a single server.

(23) The communication network **180** may facilitate communication within the system **100**. This disclosure contemplates the communication network **180** being any suitable network operable to facilitate communication between the first user device **104**, second user device **106**, real-world server **130**, virtual-world server **150** and blockchain network **190**. Communication network **180** may include any interconnecting system capable of transmitting audio, video, signals, data, messages, or any combination of the preceding. Communication network **180** may include all or a portion of a local area network (LAN), a wide area network (WAN), an overlay network, a software-defined network (SDN), a virtual private network (VPN), a packet data network (e.g., the Internet), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a Plain Old Telephone (POT) network, a wireless data network (e.g., WiFi, WiGig, WiMax, etc.), a Long Term Evolution (LTE) network, a Universal Mobile Telecommunications System (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a Near Field Communication network, a Zigbee network, and/or any other suitable network, operable to facilitate communication between the components of system **100**. In other embodiments, system **100** may not have all of these components and/or may have other elements instead of, or in addition to, those above.

(24) Each of the user devices (i.e., first user device **104** and second user device **106**) may be any computing device configured to communicate with other devices, such as a server (e.g., real-world server **130**, virtual-world server **150** and/or blockchain network **190**), databases, etc. through the communication network **180**. Each of the user devices may be configured to perform specific functions described herein and interact with one or both of real-world server **130** and the virtual-world server **150**, e.g., via respective user interfaces. Each of the user devices is a hardware device that is generally configured to provide hardware and software resources to a user. Examples of a user device include, but are not limited to, a virtual reality device, an augmented reality device, a laptop, a computer, a smartphone, a tablet, a smart device, an Internet-of-Things (IoT) device, or any other suitable type of device. The user devices may comprise a graphical user interface (e.g., a display), a touchscreen, a touchpad, keys, buttons, a mouse, or any other suitable type of hardware that allows a user to view data and/or to provide inputs into the user device. Each user device may be configured to allow a user to send requests to one or both of real-world server **130** and the virtual-world server **150**, or to another user device.

(25) Example User Device

(26) FIG. 2 is a block diagram of an embodiment of the first user device **104** used by the system of FIG. 1. First user device **104** may be configured to display the virtual environment **102** (referring to FIG. 1) within a field of view of the first user **110** (referring to FIG. 1), capture biometric, sensory, and/or physical information of the first user **110** wearing the first user device **104**, and to facilitate an electronic interaction between the first user **110** and the second user **112** (referring to FIG. 1) or between the first user **110** and an entity (e.g., represented by a virtual entity in the virtual environment **102**).

(27) First user device **104** comprises a processor **202**, a memory **204**, and a display **206**. Further embodiments may include a camera **208**, a wireless communication interface **210**, a network interface **212**, a microphone **214**, a global position system (GPS) sensor **216**, and/or one or more biometric devices **218**. First user device **104** may be configured as shown or in any other suitable configuration. For example, first user device **104** may comprise one or more additional components and/or one or more shown components may be omitted.

(28) The processor **202** comprises one or more processors operably coupled to and in signal communication with memory **204**, display **206**, camera **208**, wireless communication interface **210**, network interface **212**, microphone **214**, GPS sensor **216**, and biometric devices **218**. Processor **202** is configured to receive and transmit electrical signals among one or more of memory **204**, display

206, camera **208**, wireless communication interface **210**, network interface **212**, microphone **214**, GPS sensor **216**, and biometric devices **218**. The electrical signals are used to send and receive data (e.g., images captured from camera **208**, virtual objects to display on display **206**, etc.) and/or to control or communicate with other devices. Processor **202** may be operably coupled to one or more other devices (for example, the real-world server **130** and/or virtual-world server **150** shown in FIG. 1).

(29) The processor **202** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **202** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **202** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **202** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(30) The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions to implement the function disclosed herein, such as some or all of those described with respect to FIGS. 1 and 3. For example, processor **202** may be configured to display virtual objects on display **206**, detect hand gestures, identify virtual objects selected by a detected hand gesture, capture biometric information of a user, such as first user **110**, via one or more of camera **208**, microphone **214**, and/or biometric devices **218**, and communicate via wireless communication interface **210** with the real-world server **130**, virtual-world server **150** and/or second user device **106**. In some embodiments, the function described herein is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware or electronic circuitry.

(31) The memory **204** is operable to store any of the information described with respect to FIGS. 1 and 3 along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor **202**. For example, the memory **204** may store the instructions **220**. The memory **204** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. Memory **204** is operable to store, for example, information relating to the identity of the user (e.g., at least a portion of user data **132**), instructions for performing the functions of first user device **104** described herein, and any other data or instructions. The memory **204** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(32) Display **206** is configured to present visual information to a user (for example, first user **110** in FIG. 1) in a virtual reality environment, an augmented reality environment or mixed reality environment. In other embodiments, the display **206** is configured to present visual information to the user as the virtual environment **102** (referring to FIG. 1) in real-time. In an embodiment, display **206** is a wearable optical display (e.g., glasses or a headset) configured to reflect projected images and enables a user to see through the display. For example, display **206** may comprise display units, lens, semi-transparent mirrors embedded in an eye glass structure, a visor structure, or a helmet structure. Examples of display units include, but are not limited to, a cathode ray tube (CRT) display, a liquid crystal display (LCD), a liquid crystal on silicon (LCOS) display, a light emitting diode (LED) display, an active-matrix OLED (AMOLED), an organic LED (OLED) display, a projector display, or any other suitable type of display as would be appreciated by one of

ordinary skill in the art upon viewing this disclosure. In another embodiment, display **206** is a graphical display on a user device. For example, the graphical display may be the display of a tablet or smart phone configured to display virtual environment **102**.

(33) Examples of camera **208** include, but are not limited to, charge-coupled device (CCD) cameras and complementary metal-oxide semiconductor (CMOS) cameras. Camera **208** is configured to capture images of a wearer of first user device **104**, such as first user **110**. Camera **208** may be configured to capture images continuously, at predetermined intervals, or on-demand. For example, camera **208** may be configured to receive a command from first user **110** to capture an image. In another example, camera **208** is configured to continuously capture images to form a video stream. Camera **208** is communicably coupled to processor **202**.

(34) Examples of wireless communication interface **210** include, but are not limited to, a Bluetooth interface, an RFID interface, a near field communication interface, a local area network (LAN) interface, a personal area network interface, a wide area network (WAN) interface, a Wi-Fi interface, a ZigBee interface, or any other suitable wireless communication interface as would be appreciated by one of ordinary skill in the art upon viewing this disclosure. Wireless communication interface **210** is configured to facilitate processor **202** in communicating with other devices. For example, wireless communication interface **210** is configured to enable processor **202** to send and receive signals with other devices, such as second user device **106**, the real-world server **130** and/or virtual-world server **150** (referring to FIG. **1**). Wireless communication interface **210** is configured to employ any suitable communication protocol.

(35) The network interface **212** is configured to enable wired and/or wireless communications. The network interface **212** is configured to communicate data between the first user device **104** and other network devices, systems, or domain(s). For example, the network interface **212** may comprise a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. The processor **202** is configured to send and receive data using the network interface **212**. The network interface **212** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(36) Microphone **214** is configured to capture audio signals (e.g., voice signals or commands) from a user, such as first user **110**. Microphone **214** is configured to capture audio signals continuously, at predetermined intervals, or on-demand. Microphone **214** is communicably coupled to processor **202**.

(37) GPS sensor **216** is configured to capture and to provide geographical location information. For example, GPS sensor **216** is configured to provide a geographic location of a user, such as first user **110**, employing first user device **104**. GPS sensor **216** may be configured to provide the geographic location information as a relative geographic location or an absolute geographic location. GPS sensor **216** may provide the geographic location information using geographic coordinates (i.e., longitude and latitude) or any other suitable coordinate system. GPS sensor **216** is communicably coupled to processor **202**.

(38) Examples of biometric devices **218** may include, but are not limited to, retina scanners, fingerprint scanners and facial scanners. Biometric devices **218** are configured to capture information about a person's physical characteristics and to output a biometric signal based on captured information. A biometric signal is a signal that is uniquely linked to a person based on their physical characteristics. For example, biometric device **218** may be configured to perform a retinal scan of the user's eye and to generate a biometric signal for the user based on the retinal scan. As another example, a biometric device **218** is configured to perform a fingerprint scan of the user's finger and to generate a biometric signal for the user based on the fingerprint scan. Biometric device **218** is communicably coupled to processor **202**.

(39) Referring back to FIG. **1**, in one or more embodiments, one or both of the real-world server **130** and the virtual-world server **150**, blockchain network **190** and one or more user devices (e.g., second user device **106**) may be part of an Information Technology (IT) infrastructure of an entity

or organization. For example, second user **112** may be a representative of the organization who may use the second user device **106** to enter the virtual environment **102** and virtually interact with one or more users (e.g., first user **110**) via the second avatar **116** to provide services to the first user **110**. (40) The real-world server **130** may be configured to allow users (e.g., first user **110**) registered with the real-world server **130** to perform one or more data interactions in the real-world environment. Similarly, virtual-world server **150** may be configured to allow users (e.g., first user **110**) registered with the virtual-world server **150** to perform one or more data interactions in the virtual environment **102** (e.g., a metaverse environment). In one embodiment, the real-world server **130** and the virtual-world server **150** are owned and/or operated by the same entity/organization. In this context, virtual-world server **150** may be configured to allow users (e.g., first user **110**) registered with the real-world server **130** to perform one or more data interactions in the virtual environment **102** (e.g., a metaverse environment). In alternative embodiments, the real-world server **130** and the virtual-world server **150** may be owned and/or operated by different entities/organizations.

(41) In one or more embodiments, as the first user **110** initially registers with the real-world server **130** in the real-world environment, the real-world server **130** may collect several pieces of information from the user including information relating to the identity of the user such as legal name, social security number, biometrics (e.g., fingerprints, retina scans, face ID etc.), residence address, phone numbers, assets owned by the user, and copies of government issued documents (e.g., drivers permit, state identity card etc.). This information is stored by real-world server **130** as part of user data **132** of the first user **110**. In one embodiment, at least a portion of the user data **132** relating to the first user **110** collected in the real-world environment may be stored in the virtual-world server **150**. Once the identity of the first user **110** is confirmed and all other information provided by the first user **110** is verified to be correct, the real-world server **130** may generate a real-world data file **134** for the first user **110** in which the first user **110** may store real-world data objects **136** owned by the first user **110**. In one example, the first user **110** may engage in a real-world interaction with a service representative managing the real-world server **130** (e.g., physical interaction at an office location, over phone, voice chat etc.) to provide such information that can be used to register the first user **110** at the real-world server **130** and generate the real-world data file **134** of the first user **110**. In another example, the first user **110** may engage in a real-world interaction by accessing a webpage provided and managed by the real-world server **130**. Once the first user **110** initiates a registration process via the webpage, the real-world server **130** may walk the first user **110** through several steps in which the first user **110** may be asked to provide information necessary to verify the identity of the first user **110** and register the first user **110** with the real-world server **130**.

(42) Information relating to the real-world data file **134** of the first user **110** may be stored as part of the user data **132** of the first user **110**. This information may include, but is not limited to, an identity of the real-world data file **134**, amount of real-world data objects **136** stored in the real-world data file **134**, a log of data interactions conducted in relation to the real-world data file **134** and any other information relating to the real-world data file **134**.

(43) Once registered with the real-world server **130**, the real-world server **130** may allow the first user **110** to perform one or more data interactions in the real-world environment. For example, a real-world data interaction may include transferring one or more real-world data objects **136** from the real-world data file **134** of the first user **110** to a second real-world data file (not shown) of the second user **112**. Another example data interaction may include receiving one or more real-world data objects **136** in the real-world data file **134** of the first user **110** from the second real-world data file of the second user **112**. Another example data interaction may include requesting by the first user **110** transfer of real-world data objects from a data file of a second user to a user data file of a third user as part of satisfying an agreement between the first user **110** and the third user. Another example data interaction may include modifying at least a portion of the user data **132** (e.g., user

credentials to access the real-world server, phone numbers, residential address, email address, information relating to user assets etc.) stored at the real-world server **130**. It may be noted that a data interaction in accordance with embodiments of the present disclosure refers to any interaction in the real-world environment and/or virtual environment **102** that includes transfer of data between computing nodes (e.g., first user device **104**, second user device **106**, real-world server **130** and virtual-world server **150**).

(44) The first user **110** may additionally register with the virtual-world server **150**. In one embodiment, when initially registering with the virtual-world server **150**, the first user **110** may provide to the virtual-world server **150** a credential (e.g., username and password) that provides the first user **110** access to the real-world server **130**. In one embodiment, a single web page or web portal may allow the first user **110** to register with the real-world server **130** as well as the virtual-world server **150**. The first user **110** may first register with the real-world server **130** as described above and generate credentials that allow the first user **110** access to the real-world server **130** and services provided by the real-world server **130**. Once registered with the real-world server **130**, the web portal may offer the first user **110** an option to additionally register with the virtual-world server **150** which may allow the first user **110** to perform data interactions in the virtual environment **102**. Registration with the virtual-world server **150** may include generating a first user credential **152** that allows the first user **110** to sign on to the virtual-world server **150** and enter the virtual environment **102** via first avatar **114** of the first user **110**. Other users may register with the real-world server **130** and the virtual-world server **150** in a similar manner as the first user **110**. For example, second user **112** may register with the virtual-world server **150** which may include generating a second user credential **154** that allows the second user **112** to sign on to the virtual-world server **150** and enter the virtual environment **102** via second avatar **116** of the second user **112**.

(45) Once registered with the virtual-world server **150**, the first user **110** may generate a virtual data file **160** in which the first user **110** may store virtual data objects **162** owned by the first user **110**. In one or more embodiments, the virtual data file **160** of the first user **110** is associated with the real-world data file **134** of the first user **110**. For example, the virtual data file **160** is a virtual image of the real-world data file **134**, wherein the virtual data objects **162** correspond to the real-world data objects **136**. In other words, the virtual data file **160** is a virtual representation of the real-world data file **134**. In another example, the virtual data file **160** stores a portion of the real-world data objects **136** in the form of virtual data objects **162**. In another example, real-world data objects **136** may be converted to virtual data objects **162**, and vice versa. In this case, there may not be a one-to-one conversion between the real-world data objects **136** and virtual data objects **162**. For example, one real-world data object **136** may be converted to a plurality of virtual data objects **162**, wherein the conversion ratio may dynamically change from time to time.

(46) Information relating to the virtual data file **160** of the first user **110** may be stored by the virtual-world server **150**. This information may include, but is not limited to, an identity of the virtual data file **160**, amount of virtual data objects **162** stored in the virtual data file **160**, a log of virtual data interactions conducted in the virtual environment **102** in relation to the virtual data file **160** and any other information relating to the virtual data file **160**.

(47) Once registered with the virtual-world server **150**, the virtual-world server **150** may allow the first user **110** to perform one or more virtual data interactions. For example, a virtual data interaction may include transferring one or more virtual data objects **162** from the virtual data file **160** of the first user **110** to a second virtual data file (not shown) of the second user **112**. Another example data interaction may include receiving one or more virtual data objects **162** in the virtual data file **160** of the first user **110** from the second virtual data file of the second user **112**. Another example data interaction may include requesting by the first user **110** transfer of virtual data objects from a data file of a second user to a data file of a third user as part of satisfying an agreement between the first user **110** and the third user.

(48) In one or more embodiments, the virtual data file (e.g., virtual data file **160**) is a software application running on a computing node owned and/or operated by the respective user (e.g., first user **110**). For example, when the first user **110** desires to receive virtual data objects **162** from a virtual data file of the second user **112**, first user **110** may direct the second user **112** to a unique cryptographic address (e.g., public key) issued by the virtual data file **160**. In one embodiment, the virtual data file **160** may not itself store the virtual data objects **162** but may store information that points to a location of the virtual data objects **162**, for example, on a server (e.g., virtual-world server **150**). Virtual data file **160** may be web-based or hardware-based. For example, virtual data file **160** may be stored in a mobile device or a desktop computer connected to the internet. Additionally or alternatively, virtual data file **160** may be stored in a device (e.g., USB drive) that is not connected to the network **180**.

(49) In one or more embodiments, virtual-world server **150** may be configured to record virtual-world data interactions performed in the virtual environment **102** and/or real-world data interactions performed in the real-world environment. For example, virtual-world server **150** and/or one or more user devices (e.g., **104**, **106** etc.) connected to the network **180** may be part of the blockchain network **190** and virtual data interactions performed in the virtual environment **102** (e.g., by the virtual-world server **150** or users **110**, **112** etc.) may be recorded as data interaction history **194** in the blockchain network **190** (e.g., stored in the blockchain ledger **192**). As shown in FIG. 1, blockchain network **190** may be associated with a blockchain ledger **192**. Each virtual-world data interaction record stored in the data interaction history **194** may relate to a distinct virtual-world data interaction performed in the virtual environment **102** and may include information relating to the virtual-world data interaction including, but not limited to, a type of the data interaction (e.g., sending virtual data objects **162**, receiving virtual data objects **162**, updating user data **132** etc.), a date and time the data interaction was performed, an identity of a sending data file (e.g., virtual data file **160**), an identity of a receiving data file (e.g., virtual data file **160**), an amount of virtual data objects **162** transferred (e.g., sent or received), and an identity of an entity or user (e.g., second user **112**) with which the data interaction was performed. In an additional or alternative embodiment, the data interaction history **194** associated with the first user **110** may be stored as part of the user data **132** in the real-world server **130**.

(50) A blockchain generally is an open, decentralized and distributed digital ledger (e.g., blockchain ledger **192**) consisting of records called blocks that are used to record data interactions across many computing nodes (e.g., virtual-world server **150**, first user device **104**, second user device **106** etc.). Each computing node of a blockchain network (e.g., blockchain network **190**) may maintain a copy of the blockchain ledger **192**. Logically, a blockchain is a chain of blocks which contains specific information. Once recorded, the data in any given block cannot be altered retroactively without alteration of all subsequent blocks, which requires consensus of the network majority. Each computing node within the blockchain network maintains, approves, and updates new entries. The system is controlled not only by separate individuals, but by everyone within the blockchain network. Each member ensures that all records and procedures are in order, which results in data validity and security. Thus, the distributed ledger can record interactions between two parties efficiently and in a verifiable and permanent way. By design, a blockchain is resistant to modification of the data. For use as a distributed ledger, a blockchain is typically managed by a peer-to-peer network collectively adhering to a protocol for inter-node communication and validating new blocks. In the context of the present disclosure, one or more computing nodes (e.g., virtual-world server, first user device **104**, second user device **106** etc.) of the blockchain network **190** may store a copy of a blockchain ledger **192**, wherein each copy of the blockchain ledger **192** includes a copy of the blockchain associated with the blockchain network **190**.

(51) Each computing node of a blockchain network (e.g., blockchain network **190**) is configured to process new blocks generated for the blockchain and maintain a most recent copy of the blockchain in the respective ledgers. Any new interaction or activity within the blockchain network may

trigger the building of a new block of the blockchain. An interaction may include a computing node (e.g., virtual-world server, first user device **104**, second user device **106** etc.) of the blockchain network transmitting or receiving data from another computing node (e.g., virtual-world server, first user device **104**, second user device **106** etc.) of the blockchain network or from a computing node (e.g., real-world server **130**) that is not part of the blockchain network. Before a new block is added to the blockchain, it needs to be verified by a majority of the computing nodes in the blockchain network. For example, once a new block is created at a computing node belonging to the blockchain network **190**, the new block is sent to each other computing node in the blockchain network **190**. Each other computing node verifies the new block and checks whether the information stated in the new block is correct. Once approved by a majority of the computing nodes, the new block is added to the blockchain of the blockchain network **190**. Once a new block is approved for addition to the blockchain, each of the computing nodes of the blockchain network **190** may be configured to update a local copy of the blockchain persisted in the respective ledger to reflect the addition of the new block.

(52) Each block of the blockchain includes a hash of the block, a hash of the previous block, data that records one or more interactions or activities associated with the block, and a timestamp of the one or more interactions or activities recorded by the block. The data stored in each block depends on the type of blockchain. For example, the data included in each block may include information relating to the interaction recorded by the block including transmitting/receiving data, details of the data files, a copy of data received or generated as part of the interaction, identities of the sending and receiving nodes involved in the interaction etc. A hash of a block is like a fingerprint that uniquely identifies the block (and the interaction or activity recorded by the block) within the blockchain. Each hash of a block is generated based on a cryptographic hash algorithm. A blockchain network uses public-key cryptography to securely transfer data between computing nodes. Public-key cryptography uses a combination of a public key and private key to secure data in a blockchain network so that only the rightful owner of data can access the data. A public key is like an address on the blockchain to which data may be sent and recorded as belonging to that address. A private key is like a password that gives a user access to digital possessions recorded against a public key.

(53) In one or more embodiments of the present disclosure, virtual-world server **150** may be configured to facilitate collection and distribution of data objects (virtual data objects **162** and/or real-world data objects **136**) to one or more target entities. A target entity may be a user (e.g., first user **110**, second user **112** etc.) or an organization. In an embodiment, the target entity may or may not be registered with the virtual-world server **150**. In one example, the first user **110** may desire to collect virtual data objects **162** from one or more other users (e.g., second user **112**) in the virtual environment **102** for distribution to one or more target entities. The first user **110** may gain access to the virtual environment **102** (e.g., using first user device **104**) by providing the first user credential **152** of the first user **110** and may enter the virtual environment **102** (e.g., via first avatar **114**). Virtual-world server **150** may be configured to receive the first user credential **152** from the first user **110**, verify the identity of the first user **110** and authorize the first user **110** to access the virtual environment **102**. Once the first user **110** has gained access to the virtual environment **102**, the first user **110** may request (e.g., using the first user device **104**) the virtual-world server **150** to receive virtual data objects **162** from other users in the virtual environment **102** for distribution to one or more target entities. The request from the first user **110** may include information relating to the one or more target entities including identities of the one or more target entities that are to receive virtual data objects **162** received from users based on virtual data interactions conducted by the users within the virtual environment **102**. In response to receiving the request from the first user **110**, virtual-world server **150** may be configured to generate a virtual sub-environment **120** which other users of the virtual environment **102** may access to transfer virtual data objects **162** for distribution to the one or more target entities.

(54) A virtual sub-environment **120** may be a designated region within the virtual environment **102** for use by a user (e.g., first user **110**) or an entity/organization. The virtual sub-environment **120** may take any form in the virtual environment **102** including, but not limited to, a virtual room, a virtual building or portions thereof, a virtual store, or any other designated virtual space/region within the virtual environment **102**. The virtual sub-environment **120** is generated for a customized purpose such as for receiving virtual data objects **162** from users for distribution to specified one or more target entities and allows users (e.g., first user **110**) of the virtual environment **102** to perform virtual data interactions including transferring virtual data objects **162** for distribution to the specified one or more target entities. Virtual-world server **150** may be configured to generate different virtual sub-environments **120** for different customized purposes. For example, based on a first request from a first user, virtual-world server **150** may generate a first virtual sub-environment for collection and distribution of virtual data objects to a first set of target entities. Further, based on a second request from a second user, virtual-world server **150** may generate a second virtual sub-environment for collection and distribution of virtual data objects to a second set of target entities. Users of the virtual environment **102** may use the first and/or second virtual sub-environments to transfer virtual data objects to the respective first and/or second set of target entities. In other words, virtual data objects received from users in the first virtual sub-environment are transferred to the first set of target entities and the virtual data objects received from users in the second virtual sub-environment are transferred to the second set of target entities.

(55) The virtual sub-environment **120** may have restricted access such that only users who are registered to access the virtual sub-environment **120** may access the virtual sub-environment **120**, while other users that have access to the virtual environment **102** but are not registered to access the virtual sub-environment **120** within the virtual environment **102** may not access the virtual sub-environment **120**. In this context, it is important to have a mechanism that can help ensure that only authorized users can access the virtual sub-environment **120**.

(56) In some embodiments, gaining access to a virtual sub-environment **120** may include multiple steps. First, the second user **112** gains access to the virtual environment **102** using, for example, second user credential **154**, as described below. Subsequently, the second user **112** gains access to the virtual sub-environment **120** within the virtual environment **102** using a virtual security token **156**, as described below. In one or more embodiments, virtual-world server **150** may be configured to use a second user credential **154** collected from the second user **112**, generated by the second user **112** or assigned to the second user **112** during real-world data interactions with the second user **112**, to verify identity of the second user **112** in the virtual environment **102**. Thus, the second user credential **154** provides the second user **112** access to the virtual environment **102**. For example, the second user credential **154** may be used by the virtual-world server **150** to verify that the second avatar **116** belongs to and is controlled by the second user **112**. For example, second user credential **154** may include a username and password generated by the second user **112** as part of registering with the real-world server **130** and/or virtual-world server **150**. The virtual-world server **150** may allow the second user **112** to use the same username and password to enter the virtual environment **102** via second avatar **116**.

(57) Virtual-world server **150** may be configured to generate a virtual security token **156**, that provides the second user **112** access to the virtual sub-environment **120** within the virtual environment **102**. The virtual security token **156** may represent a virtual user credential that may include, but is not limited to, an encrypted keycard, a virtual token, a virtual tag or a virtual halo. In one embodiment, the virtual security token **156** includes an encrypted data file that can store information. The second user **112** may first enter the virtual environment **102** (e.g., via second avatar **116**) using the second user credential **154** and then access the virtual sub-environment **120** using the virtual security token **156**. Once the second avatar **116** of the second user **112** has accessed the virtual sub-environment **120**, second user **112** may perform one or more virtual data interactions in the virtual sub-environment **120** including transferring virtual data objects **162** for

distribution to one or more target entities associated with the virtual sub-environment **120**.

(58) The second user **112** may gain access to the virtual sub-environment **120** based on a relationship with the entity that manages the virtual environment **102**. For example, a user who is registered with the real-world server **130** and/or virtual-world server **150** may automatically qualify to access the virtual sub-environment **120**. The second user **112** may request the virtual-world server **150** to provide access to the virtual sub-environment **120**. In response to receiving the request, virtual-world server **150** may check whether the second user **112** is registered with the real-world server **130** and/or virtual-world server **150**. Upon determining that the second user **112** is registered with the real-world server **130** and/or the virtual-world server **150**, virtual-world server **150** may generate the virtual security token **156** that provides to the second user **112** access to the virtual sub-environment **120**.

(59) In some embodiments, as further described below, virtual-world server **150** may be configured to provide a user (e.g., second user **112**) access to the virtual sub-environment **120** based on approval from the first user **110** who requested setup of the virtual sub-environment **120** for distribution of data objects to one or more target entities.

(60) Virtual-world server **150** may be configured to provide the second user **112** access to the virtual sub-environment **120** based on the virtual security token **156** previously generated for the second user **112**. As described above, second user **112** may enter (e.g., using the second avatar **116**) the virtual environment **102** based on the second user credential **154**. After entering the virtual environment **102**, the second user **112** using the second avatar **116** may navigate to the virtual sub-environment **120** within the virtual environment **102** and request access to the virtual sub-environment **120**. The request for access may include the virtual security token **156** previously issued to the second user **112**. Virtual-world server **150** may receive the request for access from the second user **112** including the virtual security token **156**. Upon receiving the request from the second user **112** including the virtual security token **156**, virtual-world server **150** determines whether the second avatar **116** is authorized to access the virtual sub-environment **120**. In order to make this determination, the virtual-world server **150** first determines who (e.g., identity of the user) the virtual security token **156** was issued to. Additionally, virtual-world server **150** determines whether the avatar presenting the virtual security token **156** to request access to the virtual sub-environment **120** is associated to the user who was issued the virtual security token **156**.

(61) For example, in response to receiving the request, virtual-world server **150** may examine the user data **132** associated with the virtual security token **156**. Based on the user data associated with the virtual security token **156**, virtual-world server **150** may be configured to determine that the virtual security token **156** is associated with the second user **112**. For example, user data **132** associated with the virtual security token **156** may include an identity of the second user **112**. Based on the identity of the second user **112**, virtual-world server **150** may determine that the virtual security token **156** was generated for the second user **112**. Additionally or alternatively, based on the user data associated with the virtual security token **156**, virtual-world server **150** may be configured to verify that the second avatar **116** presenting the virtual security token **156** for gaining access to the virtual sub-environment **120** is associated with the second user **112** and not an unauthorized third party. Virtual-world server **150** provides the second user (e.g., second avatar **116**) access to the virtual sub-environment **120** upon successfully verifying the virtual security token **156** presented by the second user **112**.

(62) Virtual-world server may associate with the virtual sub-environment **120** the identities of the one or more target entities for which the virtual sub-environment **120** was generated. For example, when the first user **110** requests to receive virtual data objects **162** in the virtual environment **102** for distribution to a first set of target entities, virtual-world server **150** generates the virtual sub-environment **120** and associates with the virtual sub-environment **120** identities of one or more target entities in the first set of target entities specified by the first user **110**. Virtual-world server **150** may be configured to transfer virtual data objects **162** received from users, based on data

interactions conducted by the users from within the virtual sub-environment **120**, to one or more target entities of the first set of target entities associated with the virtual sub-environment **120**. For example, when the second user **112** gains access to the virtual sub-environment **120** as described above and transfers virtual data objects **162** in the virtual sub-environment **120**, virtual-world server **150** automatically transfers the virtual data objects received from the second user **112**, based on data interactions conducted by the user in the virtual sub-environment **120**, to the one or more target entities of the first set of target entities associated with the virtual sub-environment **120**. In one embodiment, the virtual data objects transferred by the second user **112** from within the virtual sub-environment **120** are stored in the virtual data file **160** of the first user **110**. The virtual data objects are then distributed to the one or more target entities from the virtual data file **160** of the first user **110**. In an alternative embodiment, virtual-world server **150** may generate a new virtual data file **160** and link the new virtual data file **160** with the virtual sub-environment **120**. Virtual data objects transferred by users from within the virtual sub-environment **120** are automatically stored in the virtual data file **160** linked to the virtual sub-environment **120** and eventually distributed to the one or more target entities.

(63) Virtual-world server **150** may be configured to transfer virtual data objects **162** received from the second user **112** to one or more target entities associated with the virtual sub-environment **120** based on one or more conditions **158** specified by the second user **112**. For example, after gaining access to the virtual sub-environment **120**, second user **112** may specify one or more conditions **158** relating to how virtual data objects received from the second user **112** are to be distributed to the one or more target entities associated with the virtual sub-environment **120**. For example, the one or more conditions **158** may include a first condition specifying that a first portion of the virtual data objects received from the second user **112** is to be transferred to a first target entity of the one or more target entities associated with the virtual sub-environment **120**. Further, the one or more conditions **158** may include a second condition specifying that a second portion of the virtual data objects received from the second user **112** is to be transferred to a second target entity of one or more target entities associated with the virtual sub-environment **120**. For example, the second user **112** may specify that half of the virtual data objects received from the second user **112** is to be transferred to the first target entity and the remaining half is to be transferred to the second target entity.

(64) Virtual-world server **150** may forward the one or more conditions **158** specified by the second user **112** to the first user **110** for approval. Upon receiving an approval from the first user **110** of the one or more conditions **158**, virtual-world server **150** may notify the second user **112** of the approval by the first user **110**. Once the one or more conditions **158** specified by the second user **112** are approved by the first user **110**, virtual-world server **150** may be configured to distribute virtual data objects received from the second user **112** to the one or more target entities associated with the virtual sub-environment **120** based on the approved conditions **158**. Following the example from the previous paragraph, virtual-world server **150** may transfer a first portion of virtual data objects received from the second user **112** to the first target entity and may transfer a second portion to the second target entity. Once the virtual data objects received from the second user **112** have been transferred to the target entities based on the conditions **158** specified by the second user **112**, virtual-world server **150** notifies the second user **112** that the virtual data objects of the second user **112** have been transferred to the intended target entities based on the conditions **158** specified by the second user **112**. For example, virtual-world server **150** sends a first indication to the second user **112** that the first portion of the data objects were transferred to the first target entity and sends a second indication to the second user **112** that the second portion of the data objects were transferred to the second target entity. The indication to the second user **112** may include a confirmation document from the respective first and second target entities including an amount of virtual data objects received by the respective target entities. Thus, the indication from the virtual-world server **150** serves as a confirmation for the second user **112** that the virtual data

objects of the second user **112** were distributed to the target entities in accordance with the conditions **158** specified by the second user **112**.

(65) In one embodiment, the first user **110** may modify the one or more conditions **158** specified by the second user **112** and forward the modified conditions **158** to the virtual-world server **150**. Virtual-world server **150** may forward the modified conditions **158** back to the second user **112**. Upon receiving an acceptance of the modified conditions **158** from the second user **112**, virtual-world server **150** applies the accepted modified conditions **158** to subsequent distributions of virtual data objects to the target entities.

(66) In one embodiment, the second user **112** may request (e.g., using second user device **106**) the virtual-world server **150** access to the virtual sub-environment **120**, wherein the request for access includes the one or more conditions **158** relating to how virtual data objects received from the second user **112** are to be distributed to the one or more target entities associated with the virtual sub-environment **120**. Upon receiving the access request from the second user **112**, virtual-world server **150** forwards the one or more conditions specified by the second user **112** to the first user **110**. The virtual-world server **150** provides the second user **112** access to the virtual sub-environment **120** only after the first user **110** approves the conditions **158** specified by the second user **112**. For example, in response to receiving an approval of the conditions **158** specified by the second user **112**, virtual-world server **150** generates the virtual security token **156** that provides the second user **112** access to the virtual sub-environment.

(67) In certain embodiments, the conditions **158** specified by the second user **112** may include a condition that specifies that if the virtual data objects received from the second user **112** are not transferred to one or more target entities based on one or more conditions specified by the second user **112**, then at least a portion of the virtual data objects are to be transferred back to the second user **112**. For example, when the virtual-world server **150** detects that the first entity did not receive a first portion of the virtual data objects in accordance with conditions **158** specified by the second user **112**, virtual-world server **150** transfers back the first portion of the virtual data objects (or all virtual data objects) to the second user **112**.

(68) It may be noted that different users accessing the virtual sub-environment **120** and transferring respective sets of virtual data objects in the virtual sub-environment **120** may specify different sets of conditions **158** based on which respective virtual data objects are to be distributed to the target entities. The virtual-world server may be configured to distribute virtual data objects **162** received from each user based on conditions **158** specified by the respective user.

(69) Virtual-world server **150** may be configured to convert virtual data objects **162** received from users in the virtual sub-environment into real-world data objects **136**. The real-world server **130** may be configured to transfer the converted real-world data objects **136** to the one or more target entities associated with the virtual sub-environment in the real-world environment. For example, virtual data objects **162** received from the second user **112** in the virtual sub-environment **120** may be converted into an equivalent amount of real-world data objects **136**. Virtual-world server **150** may transfer the converted real-world data objects **136** to the target entities based on one or more conditions **158** specified by the second user. Following the example from the previous paragraphs, virtual-world server **150** may transfer to the first target entity a first amount of the converted real-world data objects **136** that is equivalent to the first portion of the corresponding virtual data objects **162**. Additionally, virtual-world server **150** may transfer to the second target entity a second amount of the converted real-world data objects **136** that is equivalent to the second portion of the corresponding virtual data objects **162**.

(70) In certain embodiment, real-world server **130** may be configured to send to the virtual-world server **150** a confirmation of the transfer of real-world data objects **136** to the target entities in the real-world environment. The virtual-world server **150** may be configured to store in the blockchain ledger **192** the confirmations received from the real-world server **130**. Storing information relating to the real-world data interactions relating to transfer of the real-world data objects **136** to the target

entities enhances data security of the system **100** by storing these real-world data interactions in a permanent and transparent manner.

(71) In a banking use case, the system and methods disclosed in accordance with embodiments of the present disclosure may allow a first user **110** to setup a virtual funding account (e.g., crowd funding account, charity account etc.) for one or more target entities in the virtual environment **102**. As part of setting up the virtual funding account the virtual-world server **150** may generate the virtual sub-environment **120** in which other users (e.g., second user **112**) may contribute digital funds for distribution to the one or more target entities associated with the virtual sub-environment. Digital funds collected from users in the virtual environment **102** may be transferred to the target entities in the virtual environment **102** or may be transferred as real-world currency in the real-world environment. In this context, the real-world server **130** may be owned and/or operated by the bank. The virtual-world server **150** may be operated by the same bank or may be operated by another entity. The real-world data file **134** may correspond to a real-world bank account of the first user **110** and the real-world data objects **136** may correspond to the real-world funds in the bank account of the first user **110**. Similarly, virtual data file **160** may correspond to a digital wallet of the first user **110** and the virtual data objects **162** may correspond to digital currency.

(72) FIG. 3 illustrates a flowchart of an example method **300** for performing data interactions in a virtual environment **102**, in accordance with one or more embodiments of the present disclosure. Method **300** may be performed by the virtual-world server **150** shown in FIG. 1.

(73) At operation **302**, virtual-world server **150** receives a first user credential **152** associated with a first user **110**, wherein the first user credential **152** provides the first user **110** access to a virtual environment **102**.

(74) At operation **304**, virtual-world server **150** checks, based on the first user credential **152**, whether the first user **110** is authorized to access the virtual environment **102**. If the first user **110** is found to be not authorized to access the virtual environment **102**, method **300** proceeds to operation **306** wherein virtual-world server **150** denies the first user **110** access to the virtual environment **102**. On the other hand, if the first user **110** is found to be authorized to access the virtual environment **102**, method **300** proceeds to operation **308**, where the virtual-world server **150** authorizes, a first avatar **114** of the first user **110** to enter the virtual environment **102**.

(75) As described above, virtual-world server **150** may be configured to use a first user credential **152** collected from the first user **110**, generated by the first user **110** or assigned to the first user **110** during real-world data interactions with the first user **110**, to verify identity of the first user **110** in the virtual environment **102**. Thus, the first user credential **152** provides the first user **110** access to the virtual environment **102**. For example, the first user credential **152** may be used by the virtual-world server **150** to verify that the first avatar **114** belongs to and is controlled by the first user **110**.

(76) In one example, first user credential **152** may include a username and password generated by the first user **110** as part of registering with the real-world server **130**. The virtual-world server **150** may allow the first user **110** to use the same username and password to enter the virtual environment **102** via first avatar **114**. In an alternative embodiment, the first user **110** may generate a separate user credential (different from the first user credential **152**) as part of registering with the real-world server **130** and use the separate user credential to log into the real-world server **130** and perform real-world data interaction. The first user **110** may separately generate first user credential **152** as part of registering with the virtual-world server **150** (e.g., during a real-world data interaction with the real-world server **130**). Once the first user **110** has registered with the virtual-world server **150** and generated the first user credential **152**, the first user **110** may use the first user credential **152** to log into the virtual-world server **150**, gain access to the virtual environment **102** and perform virtual-world data interactions in within the virtual environment **102**.

(77) At operation **310**, virtual-world server **150** receive a first request from the first user **110** to receive virtual data objects **162** from one or more other users (e.g., second user **112**) in the virtual environment **102**, wherein the first request comprises information relating to at least one target

entity that is to receive the virtual data objects **162** that are received from the one or more other users.

(78) As described above, virtual-world server **150** may be configured to facilitate collection and distribution of data objects (virtual data objects **162** and/or real-world data objects **136**) to one or more target entities. A target entity may be a user (e.g., first user **110**, second user **112** etc.) or an organization. In an embodiment, the target entity may or may not be registered with the virtual-world server **150**. In one example, the first user **110** may desire to collect virtual data objects **162** from one or more other users (e.g., second user **112**) in the virtual environment **102** for distribution to one or more target entities. The first user **110** may gain access to the virtual environment **102** (e.g., using first user device **104**) by providing the first user credential **152** of the first user **110** and may enter the virtual environment **102** (e.g., via first avatar **114**). Virtual-world server **150** may be configured to receive the first user credential **152** from the first user **110**, verify the identity of the first user **110** and authorize the first user **110** to access the virtual environment **102**. Once the first user **110** has gained access to the virtual environment **102**, the first user **110** may request (e.g., using the first user device **104**) the virtual-world server **150** to receive virtual data objects **162** from other users in the virtual environment **102** for distribution to one or more target entities. The request from the first user **110** may include information relating to the one or more target entities including identities of the one or more target entities that are to receive virtual data objects **162** received from users based on virtual data interactions conducted by the users within the virtual environment **102**.

(79) At operation **312**, in response to receiving the first request, virtual-world server **150** generates a virtual sub-environment (e.g., virtual sub-environment **120**) within the virtual environment **102**, wherein the first virtual sub-environment is associated with the at least one target entity that is to receive the virtual data objects **162**.

(80) As described above, in response to receiving the request from the first user **110**, virtual-world server **150** may be configured to generate a virtual sub-environment **120** which other users of the virtual environment **102** may access to transfer virtual data objects **162** for distribution to the one or more target entities.

(81) A virtual sub-environment **120** may be a designated region within the virtual environment **102** for use by a user (e.g., first user **110**) or an entity/organization. The virtual sub-environment **120** may take any form in the virtual environment **102** including, but not limited to, a virtual room, a virtual building or portions thereof, a virtual store, or any other designated virtual space/region within the virtual environment **102**. The virtual sub-environment **120** is generated for a customized purpose such as for receiving virtual data objects **162** from users for distribution to specified one or more target entities and allows users (e.g., first user **110**) of the virtual environment **102** to perform virtual data interactions including transferring virtual data objects **162** for distribution to the specified one or more target entities. Virtual-world server **150** may be configured to generate different virtual sub-environments **120** for different customized purposes. For example, based on a first request from a first user, virtual-world server **150** may generate a first virtual sub-environment for collection and distribution of virtual data objects to a first set of target entities. Further, based on a second request from a second user, virtual-world server **150** may generate a second virtual sub-environment for collection and distribution of virtual data objects to a second set of target entities. Users of the virtual environment **102** may use the first and/or second virtual sub-environments to transfer virtual data objects to the respective first and/or second set of target entities. In other words, virtual data objects received from users in the first virtual sub-environment are transferred to the first set of target entities and the virtual data objects received from users in the second virtual sub-environment are transferred to the second set of target entities.

(82) At operation **314**, virtual-world server **150** authorizes a second user **112** to access the virtual sub-environment within the virtual environment **102**.

(83) As described above, virtual sub-environment **120** may have restricted access such that only

users who are registered to access the virtual sub-environment **120** may access the virtual sub-environment **120**, while other users that have access to the virtual environment **102** but are not registered to access the virtual sub-environment **120** within the virtual environment **102** may not access the virtual sub-environment **120**. In this context, it is important to have a mechanism that can help ensure that only authorized users can access the virtual sub-environment **120**.

(84) In some embodiments, gaining access to a virtual sub-environment **120** may include multiple steps. First, the second user **112** gains access to the virtual environment **102** using, for example, second user credential **154**, as described below. Subsequently, the second user **112** gains access to the virtual sub-environment **120** within the virtual environment **102** using a virtual security token **156**, as described below. In one or more embodiments, virtual-world server **150** may be configured to use a second user credential **154** collected from the second user **112**, generated by the second user **112** or assigned to the second user **112** during real-world data interactions with the second user **112**, to verify identity of the second user **112** in the virtual environment **102**. Thus, the second user credential **154** provides the second user **112** access to the virtual environment **102**. For example, the second user credential **154** may be used by the virtual-world server **150** to verify that the second avatar **116** belongs to and is controlled by the second user **112**. For example, second user credential **154** may include a username and password generated by the second user **112** as part of registering with the real-world server **130** and/or virtual-world server **150**. The virtual-world server **150** may allow the second user **112** to use the same username and password to enter the virtual environment **102** via second avatar **116**.

(85) Virtual-world server **150** may be configured to generate a virtual security token **156**, that provides the second user **112** access to the virtual sub-environment **120** within the virtual environment **102**. The virtual security token **156** may represent a virtual user credential that may include, but is not limited to, an encrypted keycard, a virtual token, a virtual tag or a virtual halo. In one embodiment, the virtual security token **156** includes an encrypted data file that can store information. The second user **112** may first enter the virtual environment **102** (e.g., via second avatar **116**) using the second user credential **154** and then access the virtual sub-environment **120** using the virtual security token **156**. Once the second avatar **116** of the second user **112** has accessed the virtual sub-environment **120**, second user **112** may perform one or more virtual data interactions in the virtual sub-environment **120** including transferring virtual data objects **162** for distribution to one or more target entities associated with the virtual sub-environment **120**.

(86) The second user **112** may gain access to the virtual sub-environment **120** based on a relationship with the entity that manages the virtual environment **102**. For example, a user who is registered with the real-world server **130** and/or virtual-world server **150** may automatically qualify to access the virtual sub-environment **120**. The second user **112** may request the virtual-world server **150** to provide access to the virtual sub-environment **120**. In response to receiving the request, virtual-world server **150** may check whether the second user **112** is registered with the real-world server **130** and/or virtual-world server **150**. Upon determining that the second user **112** is registered with the real-world server **130** and/or the virtual-world server **150**, virtual-world server **150** may generate the virtual security token **156** that provides to the second user **112** access to the virtual sub-environment **120**.

(87) In some embodiments, virtual-world server **150** may be configured to provide a user (e.g., second user **112**) access to the virtual sub-environment **120** based on approval from the first user **110** who requested setup of the virtual sub-environment **120** for distribution of data objects to one or more target entities.

(88) Virtual-world server **150** may be configured to provide the second user **112** access to the virtual sub-environment **120** based on the virtual security token **156** previously generated for the second user **112**. As described above, second user **112** may enter (e.g., using the second avatar **116**) the virtual environment **102** based on the second user credential **154**. After entering the virtual environment **102**, the second user **112** using the second avatar **116** may navigate to the virtual sub-

environment **120** within the virtual environment **102** and request access to the virtual sub-environment **120**. The request for access may include the virtual security token **156** previously issued to the second user **112**. Virtual-world server **150** may receive the request for access from the second user **112** including the virtual security token **156**. Upon receiving the request from the second user **112** including the virtual security token **156**, virtual-world server **150** determines whether the second avatar **116** is authorized to access the virtual sub-environment **120**. In order to make this determination, the virtual-world server **150** first determines who (e.g., identity of the user) the virtual security token **156** was issued to. Additionally, virtual-world server **150** determines whether the avatar presenting the virtual security token **156** to request access to the virtual sub-environment **120** is associated to the user who was issued the virtual security token **156**.

(89) For example, in response to receiving the request, virtual-world server **150** may examine the user data **132** associated with the virtual security token **156**. Based on the user data associated with the virtual security token **156**, virtual-world server **150** may be configured to determine that the virtual security token **156** is associated with the second user **112**. For example, user data **132** associated with the virtual security token **156** may include an identity of the second user **112**. Based on the identity of the second user **112**, virtual-world server **150** may determine that the virtual security token **156** was generated for the second user **112**. Additionally or alternatively, based on the user data associated with the virtual security token **156**, virtual-world server **150** may be configured to verify that the second avatar **116** presenting the virtual security token **156** for gaining access to the virtual sub-environment **120** is associated with the second user **112** and not an unauthorized third party. Virtual-world server **150** provides the second user (e.g., second avatar **116**) access to the virtual sub-environment **120** upon successfully verifying the virtual security token **156** presented by the second user **112**.

(90) At operation **316**, virtual-world server **150** receives a set of conditions **158** from the second user specifying how a first set of virtual data objects **162** of the second user **112** are to be distributed to the at least one target entity associated with the virtual sub-environment.

(91) As described above, virtual-world server **150** may be configured to transfer virtual data objects **162** received from the second user **112** to one or more target entities associated with the virtual sub-environment **120** based on one or more conditions **158** specified by the second user **112**. For example, after gaining access to the virtual sub-environment **120**, second user **112** may specify one or more conditions **158** relating to how virtual data objects received from the second user **112** are to be distributed to the one or more target entities associated with the virtual sub-environment **120**. For example, the one or more conditions **158** may include a first condition specifying that a first portion of the virtual data objects received from the second user **112** is to be transferred to a first target entity of the one or more target entities associated with the virtual sub-environment **120**. Further, the one or more conditions **158** may include a second condition specifying that a second portion of the virtual data objects received from the second user **112** is to be transferred to a second target entity of one or more target entities associated with the virtual sub-environment **120**. For example, the second user **112** may specify that half of the virtual data objects received from the second user **112** is to be transferred to the first target entity and the remaining half is to be transferred to the second target entity.

(92) At operation **318**, virtual-world server **150** receives from the first user **110** an approval of the set of conditions **158** received from the second user **112**.

(93) As described above, virtual-world server **150** may forward the one or more conditions **158** specified by the second user **112** to the first user **110** for approval. Upon receiving an approval from the first user **110** of the one or more conditions **158**, virtual-world server **150** may notify the second user **112** of the approval by the first user **110**. Once the one or more conditions **158** specified by the second user **112** are approved by the first user **110**, virtual-world server **150** may be configured to distribute virtual data objects received from the second user **112** to the one or more target entities associated with the virtual sub-environment **120** based on the approved conditions **158**. Following

the example from the previous paragraph, virtual-world server **150** may transfer a first portion of virtual data objects received from the second user **112** to the first target entity and may transfer a second portion to the second target entity. Once the virtual data objects received from the second user **112** have been transferred to the target entities based on the conditions **158** specified by the second user **112**, virtual-world server **150** notifies the second user **112** that the virtual data objects of the second user **112** have been transferred to the intended target entities based on the conditions **158** specified by the second user **112**. For example, virtual-world server **150** sends a first indication to the second user **112** that the first portion of the data objects were transferred to the first target entity and sends a second indication to the second user **112** that the second portion of the data objects were transferred to the second target entity. The indication to the second user **112** may include a confirmation document from the respective first and second target entities including an amount of virtual data objects received by the respective target entities. Thus, the indication from the virtual-world server **150** serves as a confirmation for the second user **112** that the virtual data objects of the second user **112** were distributed to the target entities in accordance with the conditions **158** specified by the second user **112**.

(94) In one embodiment, the first user **110** may modify the one or more conditions **158** specified by the second user **112** and forward the modified conditions **158** to the virtual-world server **150**.

Virtual-world server **150** may forward the modified conditions **158** back to the second user **112**.

Upon receiving an acceptance of the modified conditions **158** from the second user **112**, virtual-world server **150** applies the accepted modified conditions **158** to subsequent distributions of virtual data objects to the target entities.

(95) In one embodiment, the second user **112** may request (e.g., using second user device **106**) the virtual-world server **150** access to the virtual sub-environment **120**, wherein the request for access includes the one or more conditions **158** relating to how virtual data objects received from the second user **112** are to be distributed to the one or more target entities associated with the virtual sub-environment **120**. Upon receiving the access request from the second user **112**, virtual-world server **150** forwards the one or more conditions specified by the second user **112** to the first user **110**. The virtual-world server **150** provides the second user **112** access to the virtual sub-environment **120** only after the first user **110** approves the conditions **158** specified by the second user **112**. For example, in response to receiving an approval of the conditions **158** specified by the second user **112**, virtual-world server **150** generates the virtual security token **156** that provides the second user **112** access to the virtual sub-environment.

(96) At operation **320**, virtual-world server **150** receives the first set of virtual data objects **162** from the second user **112**, for example, based on a data interaction conducted by the second user **112** within the virtual sub-environment.

(97) At operation **322**, virtual-world server **150** transfers the first set of virtual data objects **162** to the at least one target entity based on the set of conditions **158** specified by the second user **112**.

(98) As described above, virtual-world server **150** may be configured to convert virtual data objects **162** received from users in the virtual sub-environment into real-world data objects **136**. The real-world server **130** may be configured to transfer the converted real-world data objects **136** to the one or more target entities associated with the virtual sub-environment in the real-world environment.

For example, virtual data objects **162** received from the second user **112** in the virtual sub-environment **120** may be converted into an equivalent amount of real-world data objects **136**.

Virtual-world server **150** may transfer the converted real-world data objects **136** to the target entities based on one or more conditions **158** specified by the second user. Following the example from the previous paragraphs, virtual-world server **150** may transfer to the first target entity a first amount of the converted real-world data objects **136** that is equivalent to the first portion of the corresponding virtual data objects **162**. Additionally, virtual-world server **150** may transfer to the second target entity a second amount of the converted real-world data objects **136** that is equivalent to the second portion of the corresponding virtual data objects **162**.

(99) In certain embodiment, real-world server **130** may be configured to send to the virtual-world server **150** a confirmation of the transfer of real-world data objects **136** to the target entities in the real-world environment. The virtual-world server **150** may be configured to store in the blockchain ledger **192** the confirmations received from the real-world server **130**. Storing information relating to the real-world data interactions relating to transfer of the real-world data objects **136** to the target entities enhances data security of the system **100** by storing these real-world data interactions in a permanent and transparent manner.

(100) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

(101) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(102) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system comprising: a memory that stores a first user credential associated with a first user and a second user credential associated with a second user; and at least one processor coupled to the memory, and configured to: receive the first user credential associated with the first user, wherein the first user credential provides the first user access to a virtual environment; authorize, based on the first user credential, a first avatar of the first user to enter the virtual environment; receive a first request from the first user to receive virtual data objects from one or more other users in the virtual environment, wherein the first request comprises information relating to at least one target entity that is to receive the virtual data objects received from the one or more other users; in response to receiving the first request, generate a first virtual sub-environment within the virtual environment, wherein the first virtual sub-environment is associated with identities of the respective at least one target entity specified in the first request that is to receive the virtual data objects and can only be used by the one or more other users to transfer the virtual data objects for the at least one target entity; generate a virtual data file; link the virtual data file to the first virtual sub-environment, wherein the virtual data objects transferred by the one or more other users in the first virtual sub-environment are automatically stored in the virtual data file for distribution to the at least one target entity associated with the first virtual sub-environment; receive the second user credential associated with the second user, wherein the second user credential provides the second user access to the virtual environment; authorize, based on the second user credential, a second avatar of the second user to enter the virtual environment; receive a second request from the second user to access the first virtual sub-environment within the virtual environment; authorize the second avatar of the second user to access the first virtual sub-environment; receive a set of conditions from the second user specifying how a first set of virtual data objects of the second user are to be distributed

to the at least one target entity; receive from the first user an approval of the set of conditions; receive from the second user the first set of virtual data objects; store the first set of virtual data objects in the virtual data file; convert the first set of virtual data objects stored in the virtual data file to a first set of real-world data objects; and transfer the first set of real-world data objects to the at least one target entity based on the set of conditions.

2. The system of claim 1, wherein: the set of conditions comprises: a first condition specifying that a first portion of the first set of virtual data objects is to be transferred to a first target entity of the at least one target entity; and a second condition specifying that a second portion of the first set of virtual data objects is to be transferred to a second target entity of the at least one target entity.

3. The system of claim 2, wherein the at least one processor is configured to transfer the first set of real-world data objects to the at least one target entity by: transferring to the first target entity a first portion of the first set of real-world data objects that is equivalent to the first portion of the first set of virtual data objects; and transferring to the second target entity a second portion of the first set of real-world data objects that is equivalent to the second portion of the first set of virtual data objects.

4. The system of claim 3, wherein the at least one processor is further configured to: send a first indication to the second user that the first portion of the first set of virtual data objects has been transferred to the first target entity; and send a second indication to the second user that the second portion of the first set of virtual data objects has been transferred to the second target entity.

5. The system of claim 1, wherein the set of conditions comprises a condition which specifies if the first set of virtual data objects are not transferred to the at least one target entity based on at least a portion of the set of conditions, the first set of virtual data objects are to be transferred back to the second user.

6. The system of claim 5, wherein the at least one processor is further configured to: detect that the first set of data objects were not transferred to at least one target entity based on at least the portion of the set of conditions; and transfer back the first set of data objects to the second user.

7. The system of claim 1, wherein the at least one processor is further configured to: authorize a plurality of users to access the first virtual sub-environment; receive the virtual data objects from the plurality of users; receive from the first virtual sub-environment a second set of conditions from one or more of the plurality of users relating to transferring the virtual data objects received from the one or more users to the at least one target entity; convert the virtual data objects received from the plurality of users into real-world data objects; and transfer the real-world data objects corresponding to the virtual data objects received from the one or more users to the at least one target entity based on the second set of conditions received from the one or more users.

8. The system of claim 1, wherein the at least one processor is further configured to: receive a third request from a third user to receive the virtual data objects from users in the virtual environment, wherein the third request comprises information relating to at least one second target entity that is to receive the virtual data objects from the users; in response to receiving the third request, generate a second virtual sub-environment within the virtual environment, wherein the second virtual sub-environment indicates information relating to the at least one second target entity that is to receive the virtual data objects; authorize one or more users to access the second virtual sub-environment; receive virtual data objects from the one or more users; convert the virtual data objects into real-world data objects; and transfer the virtual data objects to the at least one second target entity.

9. The system of claim 1, further comprising: a blockchain network coupled to the at least one processor, wherein the at least one processor is further configured to store one or more data interactions performed by the at least one processor in a distributed ledger of the blockchain network.

10. A method for performing data interactions, comprising: receiving a first user credential associated with a first user, wherein the first user credential provides the first user access to a virtual environment; authorizing, based on the first user credential, a first avatar of the first user to enter the virtual environment; receiving a first request from the first user to receive virtual data

objects from one or more other users in the virtual environment, wherein the first request comprises information relating to at least one target entity that is to receive the virtual data objects received from the one or more other users; in response to receiving the first request, generating a first virtual sub-environment within the virtual environment, wherein the first virtual sub-environment is associated with identities of the respective at least one target entity specified in the first request that is to receive the virtual data objects and can only be used by the one or more other users to transfer the virtual data objects for the at least one target entity; generating a virtual data file; linking the virtual data file to the first virtual sub-environment, wherein the virtual data objects transferred by the one or more other users in the first virtual sub-environment are automatically stored in the virtual data file for distribution to the at least one target entity associated with the first virtual sub-environment; receiving a second user credential associated with a second user, wherein the second user credential provides the second user access to the virtual environment; authorizing, based on the second user credential, a second avatar of the second user to enter the virtual environment; receiving a second request from the second user to access the first virtual sub-environment within the virtual environment; authorizing the second avatar of the second user to access the first virtual sub-environment; receiving a set of conditions from the second user specifying how a first set of virtual data objects of the second user are to be distributed to the at least one target entity; receiving from the first user an approval of the set of conditions; receiving from the second user the first set of virtual data objects; storing the first set of virtual data objects in the virtual data file; converting the first set of virtual data objects stored in the virtual data file to a first set of real-world data objects; and transferring the first set of real-world data objects to the at least one target entity based on the set of conditions.

11. The method of claim 10, wherein: the set of conditions comprises: a first condition specifying that a first portion of the first set of virtual data objects is to be transferred to a first target entity of the at least one target entity; and a second condition specifying that a second portion of the first set of virtual data objects is to be transferred to a second target entity of the at least one target entity.

12. The method of claim 11, transferring the first set of real-world data objects to the at least one target entity comprises: transferring to the first target entity a first portion of the first set of real-world data objects that is equivalent to the first portion of the first set of virtual data objects; and transferring to the second target entity a second portion of the first set of real-world data objects that is equivalent to the second portion of the first set of virtual data objects.

13. The method of claim 12, further comprising: sending a first indication to the second user that the first portion of the first set of virtual data objects has been transferred to the first target entity; and sending a second indication to the second user that the second portion of the first set of virtual data objects has been transferred to the second target entity.

14. The method of claim 10, wherein the set of conditions comprises a condition which specifies if the first set of virtual data objects are not transferred to the at least one target entity based on at least a portion of the set of conditions, the first set of virtual data objects are to be transferred back to the second user.

15. The method of claim 14, further comprising: detecting that the first set of data objects were not transferred to at least one target entity based on at least the portion of the set of conditions; and transferring back the first set of data objects to the second user.

16. A non-transitory computer-readable medium storing instructions that when executed by a processor cause the processor to: receive a first user credential associated with a first user, wherein the first user credential provides the first user access to a virtual environment; authorize, based on the first user credential, a first avatar of the first user to enter the virtual environment; receive a first request from the first user to receive virtual data objects from one or more other users in the virtual environment, wherein the first request comprises information relating to at least one target entity that is to receive the virtual data objects received from the one or more other users; in response to receiving the first request, generate a first virtual sub-environment within the virtual environment,

wherein the first virtual sub-environment is associated with identities of the respective at least one target entity specified in the first request that is to receive the virtual data objects and can only be used by the one or more other users to transfer the virtual data objects for the at least one target entity; generate a virtual data file; link the virtual data file to the first virtual sub-environment, wherein the virtual data objects transferred by the one or more other users in the first virtual sub-environment are automatically stored in the virtual data file for distribution to the at least one target entity associated with the first virtual sub-environment; receive a second user credential associated with a second user, wherein the second user credential provides the second user access to the virtual environment; authorize, based on the second user credential, a second avatar of the second user to enter the virtual environment; receive a second request from the second user to access the first virtual sub-environment within the virtual environment; authorize the second avatar of the second user to access the first virtual sub-environment; receive a set of conditions from the second user specifying how a first set of virtual data objects of the second user are to be distributed to the at least one target entity; receive from the first user an approval of the set of conditions; receive from the second user the first set of virtual data objects; store the first set of virtual data objects in the virtual data file; convert the first set of virtual data objects stored in the virtual data file to a first set of real-world data objects; and transfer the first set of real-world data objects to the at least one target entity based on the set of conditions.

17. The non-transitory computer-readable medium of claim 16, wherein: the set of conditions comprises: a first condition specifying that a first portion of the first set of virtual data objects is to be transferred to a first target entity of the at least one target entity; and a second condition specifying that a second portion of the first set of virtual data objects is to be transferred to a second target entity of the at least one target entity.

18. The non-transitory computer-readable medium of claim 17, wherein transferring the first set of real-world data objects to the at least one target entity comprises: transferring to the first target entity a first portion of the first set of real-world data objects that is equivalent to the first portion of the first set of virtual data objects; and transferring to the second target entity a second portion of the first set of real-world data objects that is equivalent to the second portion of the first set of virtual data objects.

19. The non-transitory computer-readable medium of claim 18, wherein the instructions further cause the processor to: send a first indication to the second user that the first portion of the first set of virtual data objects has been transferred to the first target entity; and send a second indication to the second user that the second portion of the first set of virtual data objects has been transferred to the second target entity.

20. The non-transitory computer-readable medium of claim 16, wherein the set of conditions comprises a condition which specifies if the first set of virtual data objects are not transferred to the at least one target entity based on at least a portion of the set of conditions, the first set of virtual data objects are to be transferred back to the second user.
